Anàlisi de l'entrenament

Comparem els resultats del DQN i el NFSP entrenats amb 200.000 episodis. Volem veure:

1. Si els models aprenen
2. Quin aprèn més ràpid
3. El resultat d'un enfrontament directe

In [7]:
```
import sys, os
sys.path.insert(0, os.path.abspath(r'c:\Users\ferri\Documents\ProjectesCodi\TFG-tru

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import torch
import logging

logging.basicConfig(level=logging.ERROR, force=True)

BASE = r'C:\Users\ferri\Documents\ProjectesCodi\TFG-truc\entrenament\entrenamentRLC
LOG_DIR = os.path.join(BASE, 'logs')
MODEL_DIR = os.path.join(BASE, 'models')

dqn_df = pd.read_csv(os.path.join(LOG_DIR, 'dqn_log.csv'))
nfsp_df = pd.read_csv(os.path.join(LOG_DIR, 'nfsp_log.csv'))

print(f"DQN: {len(dqn_df)} registres, NFSP: {len(nfsp_df)} registres")
```

```
DQN: 100 registres, NFSP: 100 registres
```

In [8]:
```
import os
import torch
import numpy as np
from entorn import TrucEnv
from rlcard.agents import DQNAgent, NFSPAgent, RandomAgent
from entrenament.entrenamentsUnificats.entrenaments_unificats import wrap_env_aplan
import types

# Wrapper especial per models antics (343 features)
def wrap_legacy_aplanat(env):
    """Aplanem l'OBS a 343 per models antics"""
    original_extract_state = env._extract_state

    def _extract_state_patched(self, state):
        extracted = original_extract_state(state)
        obs = extracted['obs']
        # obs is {'obs_cartes': (6,4,9), 'obs_context': (17,)}
        cartes = obs['obs_cartes'].flatten() # 216
        context = obs['obs_context'] # 17

        # Pad cartes to 9 canals (324)
```

```
        cartes_padded = np.zeros(324, dtype=np.float32)
        cartes_padded[:216] = cartes

        # Pad context to 19 (com era antigament)
        context_padded = np.zeros(19, dtype=np.float32)
        context_padded[:17] = context

        flat = np.concatenate([cartes_padded, context_padded], axis=0)
        extracted['obs'] = flat
        return extracted

    env._extract_state = types.MethodType(_extract_state_patched, env)
    env.state_shape = [[343] for _ in range(env.num_jugadors)]
    return env

def crear_eval_env_legacy():
    config = {
        'num_jugadors': 2,
        'cartes_jugador': 3,
        'puntuacio_final': 12,
        'seed': 42,
        'verbose': False,
    }
    return wrap_legacy_aplanat(TrucEnv(config))

device = torch.device("cpu")
temp_env = crear_eval_env_legacy()

# Carregar models amb arquitectura 343
def carregar_dqn_legacy(path, env, device):
    agent = DQNAgent(
        num_actions=env.num_actions,
        state_shape=env.state_shape[0], # 343
        mlp_layers=[256, 256],
        device=device
    )
    agent.q_estimator.qnet.load_state_dict(torch.load(path, map_location=device))
    return AgentCongelat(agent)

def carregar_nfsp_legacy(path, env, device):
    agent = NFSPAgent(
        num_actions=env.num_actions,
        state_shape=env.state_shape[0], # 343
        hidden_layers_sizes=[256, 256],
        q_mlp_layers=[256, 256],
        device=device
    )
    weights = torch.load(path, map_location=device)
    # Note: older NFSP format might use different keys, but if it's the same script
    # Or q_net and sl_net as seen in the code earlier
    try:
        agent._rl_agent.q_estimator.qnet.load_state_dict(weights['q_net'])
        agent.policy_network.load_state_dict(weights['sl_net'])
    except:
        agent._rl_agent.q_estimator.qnet.load_state_dict(weights['q'])
        agent.policy_network.load_state_dict(weights['sl'])
```

```
    return AgentCongelat(agent)

print("Carregant models legacy (343 features)...")
dqn_best = carregar_dqn_legacy(os.path.join(MODEL_DIR, "dqn_truc_best.pt"), temp_en
nfsp_best = carregar_nfsp_legacy(os.path.join(MODEL_DIR, "nfsp_truc_p0_best.pt"), t
print("Models carregats correctament.")
```

```
Carregant models legacy (343 features)...
Models carregats correctament.
```

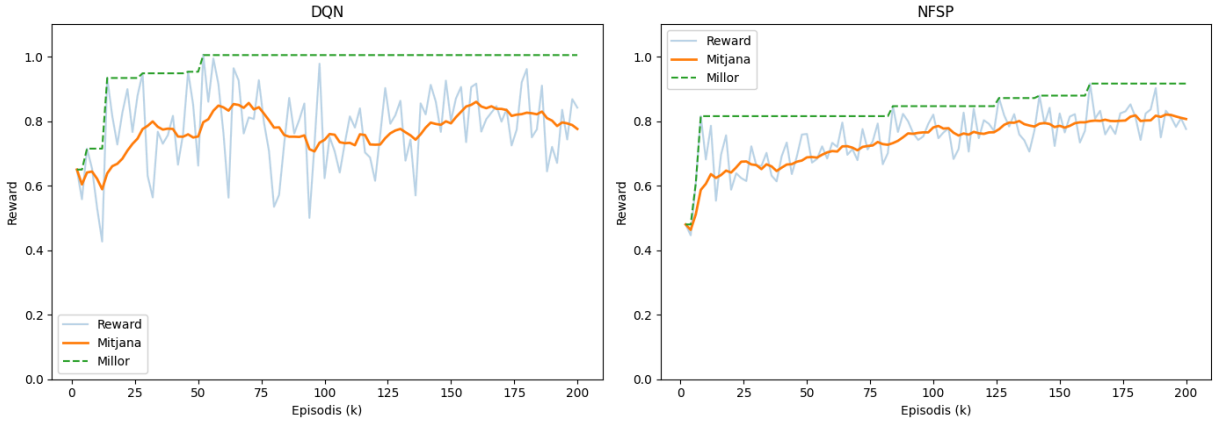
Aprenentatge

Avaluem si els models milloren. El reward va de -1 a +1. Contra un jugador aleatori, haurien d'arribar a ~0.9.

In [9]:

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# DQN
ax = axes[0]
episodes = dqn_df['episodi'] / 1000
reward = dqn_df['reward_mig']
best = dqn_df['millor_historica']

rolling = reward.rolling(window=10, min_periods=1).mean()

ax.plot(episodes, reward, alpha=0.3, label='Reward')
ax.plot(episodes, rolling, linewidth=2, label='Mitjana')
ax.plot(episodes, best, linestyle='--', label='Millor')

ax.set_xlabel('Episodis (k)')
ax.set_ylabel('Reward')
ax.set_title('DQN')
ax.legend()
ax.set_ylim(0, 1.1)

# NFSP
ax = axes[1]
episodes_n = nfsp_df['episodi'] / 1000
reward_n = nfsp_df['reward_mig_p0']
best_n = nfsp_df['millor_historica']

rolling_n = reward_n.rolling(window=10, min_periods=1).mean()

ax.plot(episodes_n, reward_n, alpha=0.3, label='Reward')
ax.plot(episodes_n, rolling_n, linewidth=2, label='Mitjana')
ax.plot(episodes_n, best_n, linestyle='--', label='Millor')

ax.set_xlabel('Episodis (k)')
ax.set_ylabel('Reward')
ax.set_title('NFSP')
ax.legend()
ax.set_ylim(0, 1.1)
```

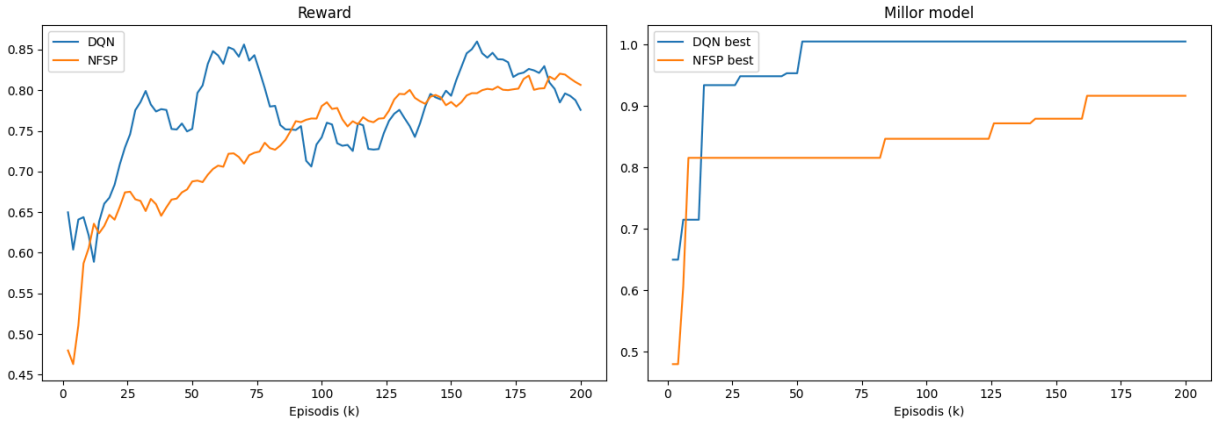
```
plt.tight_layout()
plt.show()
```



Comparació DQN vs NFSP

Sobreposem les corbes per comparar.

In [10]:
```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

episodes_k = dqn_df['episodi'] / 1000

dqn_roll = dqn_df['reward_mig'].rolling(window=10, min_periods=1).mean()
nfsp_roll = nfsp_df['reward_mig_p0'].rolling(window=10, min_periods=1).mean()

ax1.plot(episodes_k, dqn_roll, label='DQN')
ax1.plot(episodes_k, nfsp_roll, label='NFSP')
ax1.set_xlabel('Episodis (k)')
ax1.set_title('Reward')
ax1.legend()

ax2.plot(episodes_k, dqn_df['millor_historica'], label='DQN best')
ax2.plot(episodes_k, nfsp_df['millor_historica'], label='NFSP best')
ax2.set_xlabel('Episodis (k)')
ax2.set_title('Millor model')
ax2.legend()

plt.tight_layout()
plt.show()
```



In [11]:
```
dqn_rewards = dqn_df['reward_mig']
nfsp_rewards = nfsp_df['reward_mig_p0']

def stats_per_fase(rewards, episodis):
    fases = {
        '0-50k': episodis <= 50000,
        '50k-100k': (episodis > 50000) & (episodis <= 100000),
        '100k-150k': (episodis > 100000) & (episodis <= 150000),
        '150k-200k': (episodis > 150000),
    }
    rows = []
    for nom, mask in fases.items():
        r = rewards[mask]
        rows.append({
            'Fase': nom,
            'Reward': f"{r.mean():.4f}",
            'Std': f"{r.std():.4f}",
            'Max': f"{r.max():.4f}",
        })
    return pd.DataFrame(rows)

print("Estadístiques DQN")
display(stats_per_fase(dqn_rewards, dqn_df['episodi']))

print("\nEstadístiques NFSP")
display(stats_per_fase(nfsp_rewards, nfsp_df['episodi']))
```

```
Estadístiques DQN
```

	Fase	Reward	Std	Max
0	0-50k	0.7395	0.1371	0.9532
1	50k-100k	0.7903	0.1467	1.0048
2	100k-150k	0.7734	0.0928	0.9255
3	150k-200k	0.8142	0.0831	0.9617

```
Estadístiques NFSP
```

	Fase	Reward	Std	Max
0	0-50k	0.6618	0.0887	0.8155
1	50k-100k	0.7435	0.0520	0.8465
2	100k-150k	0.7799	0.0525	0.8792
3	150k-200k	0.8061	0.0453	0.9165

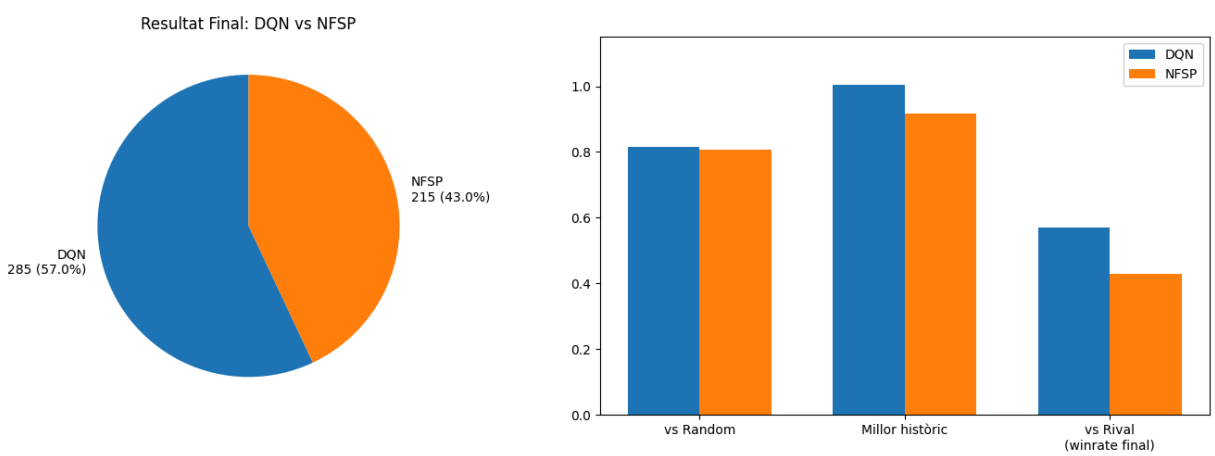
Enfrontament final

Enfrontament definitiu entre els millors models de cada algorisme. Utilitzem 500 partides alternant posicions.

In [ ]:
```
NUM_FINAL = 1000

print(f"Enfrontament final: DQN vs NFSP ({NUM_FINAL} partides)\n")

meitat = NUM_FINAL // 2
wins_dqn = 0
wins_nfsp = 0
total_reward_dqn = 0

env_final = crear_eval_env_legacy()
env_final.set_agents([dqn_best, nfsp_best])
for _ in range(meitat):
    _, payoffs = env_final.run(is_training=False)
    total_reward_dqn += payoffs[0]
    if payoffs[0] > payoffs[1]:
        wins_dqn += 1
    elif payoffs[1] > payoffs[0]:
        wins_nfsp += 1

env_final.set_agents([nfsp_best, dqn_best])
for _ in range(meitat):
    _, payoffs = env_final.run(is_training=False)
    total_reward_dqn += payoffs[1]
    if payoffs[1] > payoffs[0]:
        wins_dqn += 1
    elif payoffs[0] > payoffs[1]:
        wins_nfsp += 1

empats = NUM_FINAL - wins_dqn - wins_nfsp
wr_dqn = wins_dqn / NUM_FINAL
wr_nfsp = wins_nfsp / NUM_FINAL
reward_mig_dqn = total_reward_dqn / NUM_FINAL

print(f"  DQN:  {wins_dqn} victòries ({wr_dqn:.1%})")
print(f"  NFSP: {wins_nfsp} victòries ({wr_nfsp:.1%})")
print(f"  Empats: {empats}")
print(f"\n  Reward mig DQN: {reward_mig_dqn:.4f}")
```

```
guanyador = "DQN" if wins_dqn > wins_nfsp else ("NFSP" if wins_nfsp > wins_dqn else
print(f"\n  Guanya: {guanyador}")
```

```
Enfrontament final: DQN vs NFSP (500 partides)
  DQN:  285 victòries (57.0%)
  NFSP: 215 victòries (43.0%)
  Empats: 0

  Reward mig DQN: 0.1591

  Guanya: DQN
```

In [13]:

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

labels = [f'DQN\n{wins_dqn} ({wr_dqn:.1%})', f'NFSP\n{wins_nfsp} ({wr_nfsp:.1%})']
sizes = [wins_dqn, wins_nfsp]

ax1.pie(sizes, labels=labels, autopct='', startangle=90)
ax1.set_title('Resultat Final: DQN vs NFSP')

categories = ['vs Random', 'Millor històric', 'vs Rival\n(winrate final)']
dqn_vals = [dqn_df[dqn_df['episodi']>150000]['reward_mig'].mean(),
            dqn_df['millor_historica'].iloc[-1],
            wr_dqn]
nfsp_vals = [nfsp_df[nfsp_df['episodi']>150000]['reward_mig_p0'].mean(),
             nfsp_df['millor_historica'].iloc[-1],
             wr_nfsp]

x = np.arange(len(categories))
width = 0.35

ax2.bar(x - width/2, dqn_vals, width, label='DQN')
ax2.bar(x + width/2, nfsp_vals, width, label='NFSP')
ax2.set_xticks(x)
ax2.set_xticklabels(categories)
ax2.legend()
ax2.set_ylim(0, 1.15)

plt.tight_layout()
plt.show()
```